

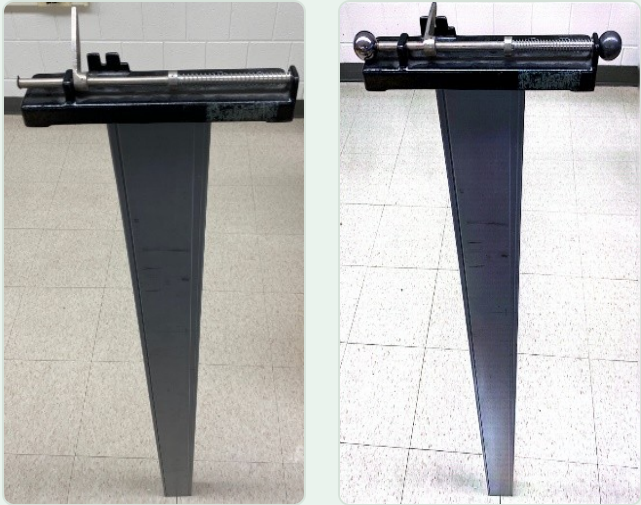
Name: _____

Date: _____

Freefall vs Horizontal Launch — How does the fall time compare?

SYNOPSIS

Imagine a cannon ball fired from the top of a tower. The cannon is pointing perfectly horizontally. At the same time as the cannon fires, a cannonball is dropped from the tower. The cannonball fired from the cannon will travel a horizontal distance before hitting the ground. A dropped cannonball will hit the ground after some time. If you measure the time it takes for the cannonballs to hit the ground, will you see a longer time for one (if so, which one), or will you see the same time (if so, why?)



PERFORMANCE EXPECTATIONS

Perform a CER exercise about the fall times of a freefalling and perfectly horizontally launched object.

Instructions and Questions

1. What is your **prediction**?

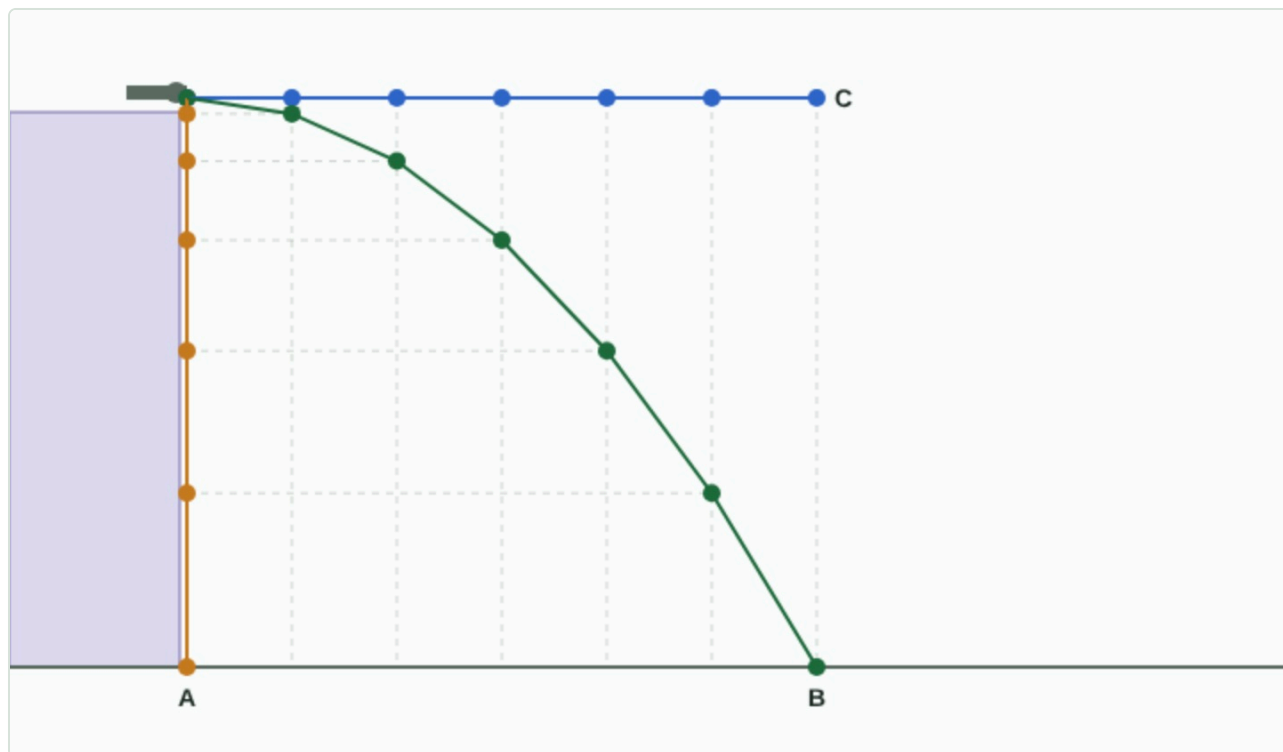
Claim — Are fall times different or the same?

Evidence — What are your previous experiences and observations related to this situation?

Reasoning — Use your knowledge of velocity, acceleration, force (importantly, the vector nature of these quantities) to justify your claim.

2. **Observe** the simulated situation using the device shown in the figure above — one end pushes a steel ball while simultaneously dropping a steel ball. How do the fall times compare?
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3. **Explain:** Given the result, revise or refine your hypothesis and reasoning. In your explanation, draw the velocity vectors in the diagram of the falling and launched balls at regular intervals below. Note that the free-falling cannonball's trajectory is A, and the horizontally launched cannonball's trajectory is B. C shows a hypothetical trajectory IF the cannonball were fired in space, where there is no gravity. Include in your explanation why Path C looks the way it is.
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Draw a velocity vector at each dot on A, B, and C, then answer the Explain question above.
